

Problem Set 2: Solutions

Labor Economics (Saggio), Summer 2026

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Question 1: Katz and Murphy (1992)

Output is produced with the CES technology

$$F_t(H_t, L_t) = \left(A_{H_t} H_t^{\frac{\sigma-1}{\sigma}} + A_{L_t} L_t^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}}, \quad \rho \equiv \frac{\sigma-1}{\sigma} \in (-\infty, 1), \quad (1)$$

so that the outer exponent is $1/\rho = \sigma/(\sigma-1)$.

Part (a): special cases

Write $G \equiv A_H H^\rho + A_L L^\rho$ and $F = G^{1/\rho}$.

$\sigma \rightarrow 0$ ($\rho \rightarrow -\infty$): **Leontief.** Suppose $H < L$, so $H^\rho > L^\rho$. Factoring the dominant term,

$$\frac{1}{\rho} \log G = \log H + \frac{1}{\rho} \log A_H + \frac{1}{\rho} \log \left(1 + \frac{A_L}{A_H} (L/H)^\rho \right) \xrightarrow{\rho \rightarrow -\infty} \log H,$$

because $(L/H)^\rho \rightarrow 0$ and $\frac{1}{\rho} \log A_H \rightarrow 0$. Hence $\boxed{F \rightarrow \min\{H_t, L_t\}}$: high- and low-skill labor are *perfect complements* and the augmenting terms drop out of the limit.

$\sigma \rightarrow \infty$ ($\rho \rightarrow 1$): **linear.** The outer exponent $1/\rho \rightarrow 1$ and $\boxed{F \rightarrow A_{H_t} H_t + A_{L_t} L_t}$: the two skill groups are *perfect substitutes*.

$\sigma \rightarrow 1$ ($\rho \rightarrow 0$): **Cobb-Douglas.** Using $A_H H^\rho = A_H e^{\rho \log H} = A_H (1 + \rho \log H + O(\rho^2))$,

$$\log F = \frac{1}{\rho} \log(A_H + A_L) + \frac{A_H \log H + A_L \log L}{A_H + A_L} + O(\rho).$$

The first term diverges unless $A_H + A_L = 1$. Under the natural Katz-Murphy normalization $A_{H_t} + A_{L_t} = 1$ (the weights are shares) it vanishes and $\boxed{F \rightarrow H_t^{A_{H_t}} L_t^{A_{L_t}}}$, Cobb-Douglas with unit elasticity of substitution.

Part (b): the skill premium with inelastic supply

Treat the quantities H_t, L_t as given (relative supply predetermined) and let input markets be competitive, so each factor is paid its marginal product. Differentiating $F = G^{1/\rho}$,

$$w_H = \frac{\partial F}{\partial H} = G^{\frac{1}{\rho}-1} A_H H^{\rho-1}, \quad w_L = \frac{\partial F}{\partial L} = G^{\frac{1}{\rho}-1} A_L L^{\rho-1}.$$

The common factor $G^{1/\rho-1}$ cancels in the ratio, and using $\rho - 1 = -1/\sigma$,

$$\frac{w_{H_t}}{w_{L_t}} = \frac{A_{H_t}}{A_{L_t}} \left(\frac{H_t}{L_t} \right)^{-1/\sigma} \implies \boxed{\log \omega_t = \log \frac{A_{H_t}}{A_{L_t}} - \frac{1}{\sigma} \log \frac{H_t}{L_t}} \quad (2)$$

This is the canonical Katz-Murphy relative-demand equation.

- Skill-biased technical change: $\frac{\partial \log \omega_t}{\partial \log(A_{H_t}/A_{L_t})} = 1 > 0$. A rise in relative skill-augmenting productivity raises the premium one-for-one.
- Relative supply: $\frac{\partial \log \omega_t}{\partial \log(H_t/L_t)} = -\frac{1}{\sigma} < 0$. More skilled labor lowers its relative price; the larger is σ (the easier the substitution), the smaller the depressing effect.

Part (c): elastic labor supply

Now $H_t = H_{0t} w_{H_t}^\epsilon$ and $L_t = L_{0t} w_{L_t}^\epsilon$, so

$$\log \frac{H_t}{L_t} = \log \frac{H_{0t}}{L_{0t}} + \epsilon \log \omega_t. \quad (3)$$

Substituting the demand relation from (b), $\log \omega_t = \log \frac{A_{H_t}}{A_{L_t}} - \frac{1}{\sigma} \log \frac{H_t}{L_t}$, into (3) and solving the resulting fixed point gives

$$\boxed{\log \omega_t = \frac{\sigma}{\sigma + \epsilon} \log \frac{A_{H_t}}{A_{L_t}} - \frac{1}{\sigma + \epsilon} \log \frac{H_{0t}}{L_{0t}}} \quad (4)$$

- SBTC: $\frac{\partial \log \omega_t}{\partial \log(A_{H_t}/A_{L_t})} = \frac{\sigma}{\sigma + \epsilon} \in (0, 1)$. The effect is *attenuated* relative to the inelastic case: a higher premium calls forth more skilled labor, which partially undoes the wage increase. As $\epsilon \rightarrow 0$ we recover 1; as $\epsilon \rightarrow \infty$ the premium becomes pinned down by supply and the effect vanishes.
- Exogenous relative supply: $\frac{\partial \log \omega_t}{\partial \log(H_{0t}/L_{0t})} = -\frac{1}{\sigma + \epsilon}$, also attenuated relative to $-1/\sigma$.

Part (d): the Katz-Murphy regression

Let $a_t \equiv \log \frac{A_{H_t}}{A_{L_t}} = \lambda_0 + \lambda_1 t + e_t$ and $s_t \equiv \log \frac{H_{0t}}{L_{0t}} = \psi_0 + \nu_t$, with $e_t \perp \nu_t$ i.i.d., $\text{var}(e_t) = V_e$, $\text{var}(\nu_t) = V_\nu$. From part (c) the equilibrium objects are

$$y_t \equiv \log \omega_t = \frac{\sigma}{\sigma + \epsilon} a_t - \frac{1}{\sigma + \epsilon} s_t, \quad x_t \equiv \log \frac{H_t}{L_t} = \frac{\epsilon \sigma}{\sigma + \epsilon} a_t + \frac{\sigma}{\sigma + \epsilon} s_t, \quad (5)$$

the second line obtained by inserting $\log \omega_t$ into (3). We run $y_t = \beta_0 + \beta_1 t + \beta_2 x_t + u_t$. Because t is included, apply Frisch-Waugh-Lovell: project everything on $(1, t)$. Since e_t, ν_t are i.i.d. (orthogonal to t), the residualized regressors are

$$\tilde{y}_t = \frac{\sigma}{\sigma + \epsilon} e_t - \frac{1}{\sigma + \epsilon} \nu_t, \quad \tilde{x}_t = \frac{\epsilon \sigma}{\sigma + \epsilon} e_t + \frac{\sigma}{\sigma + \epsilon} \nu_t.$$

With $\text{cov}(e, \nu) = 0$,

$$\text{cov}(\tilde{y}, \tilde{x}) = \frac{\sigma}{(\sigma + \epsilon)^2} (\sigma \epsilon V_e - V_\nu), \quad \text{var}(\tilde{x}) = \frac{\sigma^2}{(\sigma + \epsilon)^2} (\epsilon^2 V_e + V_\nu),$$

so that

$$\boxed{\beta_2 = \frac{\sigma\epsilon V_e - V_\nu}{\sigma(\epsilon^2 V_e + V_\nu)}} \quad (6)$$

For the trend coefficient, match the deterministic (t) parts of y_t and x_t , whose slopes are $g_y = \frac{\sigma}{\sigma+\epsilon}\lambda_1$ and $g_x = \frac{\epsilon\sigma}{\sigma+\epsilon}\lambda_1$. Since $g_y = \beta_1 + \beta_2 g_x$,

$$\beta_1 = g_y - \beta_2 g_x = \frac{\sigma\lambda_1}{\sigma+\epsilon}(1 - \epsilon\beta_2) = \frac{\sigma\lambda_1}{\sigma+\epsilon} \cdot \frac{(\sigma+\epsilon)V_\nu}{\sigma(\epsilon^2 V_e + V_\nu)},$$

which simplifies to

$$\boxed{\beta_1 = \frac{V_\nu}{\epsilon^2 V_e + V_\nu} \lambda_1} \quad (7)$$

Special cases.

Case	β_2	β_1	
(a) $\epsilon = 0$ (inelastic supply)	$-1/\sigma$	λ_1	demand identified
(b) $V_e = 0$ (deterministic tech)	$-1/\sigma$	λ_1	demand identified
(c) $V_\nu = 0$ (no supply shocks)	$1/\epsilon$	0	supply identified

Thus $\beta_2 = -1/\sigma$ and $\beta_1 = \lambda_1$ in exactly the same two cases: (a) and (b). In case (c) the regression instead traces out the *supply* curve, returning $1/\epsilon$ and a zero trend.

Intuition. The relative-wage equation is one half of a simultaneous system. OLS of $\log\omega$ on $\log(H/L)$ recovers the slope of the demand curve, $-1/\sigma$, only if the variation in relative *quantities* (after detrending) is driven by supply shifts rather than by demand (technology) shifts. That happens when (a) supply does not respond to wages ($\epsilon = 0$), so observed quantities are exogenous; or (b) technology contains no idiosyncratic shocks beyond the smooth trend ($V_e = 0$), so the trend t absorbs all of demand and the residual quantity movements are pure supply shifts that act as instruments. When instead $V_\nu = 0$ and $\epsilon, V_e > 0$, all quantity fluctuations come from demand shocks, and OLS identifies the supply slope $1/\epsilon$ with $\beta_1 = 0$.

Part (e): are the assumptions plausible?

The KM regression treats relative supply as if exogenous to the relative-wage equation, and reads $-1/\sigma$ off β_2 and the technology trend off β_1 . From (d) this is legitimate only if either $\epsilon \approx 0$ or $V_e \approx 0$:

- $\epsilon \approx 0$ is the more defensible assumption. The relative supply of college vs. high-school labor is governed by slow-moving cohort sizes and schooling decisions taken years earlier; at annual frequency it is close to *predetermined*. This is, implicitly, what KM rely on.
- $V_e = 0$ (a purely deterministic, smoothly trending demand) is less believable literally, since technology surely receives shocks, but it is a reasonable approximation if those shocks are small relative to supply movements.
- The real danger is simultaneity: if demand has genuine high-frequency shocks ($V_e > 0$) and supply responds to wages ($\epsilon > 0$), then β_2 is a variance-weighted average of $-1/\sigma$ and $1/\epsilon$ and is biased *upward* (toward zero/positive), while $\beta_1 \neq \lambda_1$.

On balance the exercise is informative if one is willing to treat relative supply as predetermined, but the inference is fragile; this is exactly why the later literature (Card-Lemieux, Krusell et al., Acemoglu-Autor) enriches the demand system and instruments relative supply.

Question 2: Skill Aggregation and Skill Prices

Output is $Q = (\sum_i A_i)^\alpha (\sum_i M_i)^\beta (\sum_i A_i M_i)^{1-\alpha-\beta}$. Hippies supply $(A, M) = (2, 1)$ and Nerds $(A, M) = (1, 3)$. With H hippies and N nerds the three aggregates are

$$\sum_i A_i = 2H + N, \quad \sum_i M_i = H + 3N, \quad \sum_i A_i M_i = 2H + 3N,$$

the last because each hippy contributes $A_i M_i = 2$ and each nerd $A_i M_i = 3$. So

$$Q = (2H + N)^\alpha (H + 3N)^\beta (2H + 3N)^{1-\alpha-\beta}. \quad (8)$$

Part (a): returns to scale and elasticity of substitution

Each aggregate is linear, hence homogeneous of degree 1 in (H, N) , and the exponents sum to $\alpha + \beta + (1 - \alpha - \beta) = 1$. Scaling $(H, N) \rightarrow (tH, tN)$ multiplies each factor by t and Q by t^1 : Q is **CRTS** in (H, N) .

For the elasticity of substitution between the two *types*, write $P = 2H + N$, $R = H + 3N$, $S = 2H + 3N$, $\gamma = 1 - \alpha - \beta$, and use that for a linearly homogeneous $Q(H, N)$, $\sigma_{HN} = \frac{Q_H Q_N}{Q Q_{HN}}$. With $\theta_H \equiv \frac{Q_H}{Q} = \frac{2\alpha}{P} + \frac{\beta}{R} + \frac{2\gamma}{S}$, $\theta_N \equiv \frac{Q_N}{Q} = \frac{\alpha}{P} + \frac{3\beta}{R} + \frac{3\gamma}{S}$, and $D \equiv \frac{2\alpha}{P^2} + \frac{3\beta}{R^2} + \frac{6\gamma}{S^2}$, one finds $Q_{HN} = Q(\theta_H \theta_N - D)$ and therefore

$$\sigma_{HN} = \frac{\theta_H \theta_N}{\theta_H \theta_N - D} \quad (9)$$

This is positive and *finite* (CRTS implies $Q_{HN} > 0$, so the denominator is positive), but it is *not constant*: it varies with the H/N mix, so Q is not CES in (H, N) even though it is built from perfect-substitute skill aggregates. Numerically, at $\alpha = \beta = \frac{1}{3}$ it ranges from $\sigma_{HN} \approx 8.3$ at $H = N$ to ≈ 14.6 at $H/N = 1/5$ (see `ojs_solve.py`). The economic content: Hippies and Nerds are close, but imperfect, substitutes, because they carry different *bundles* of the two underlying skills.

Part (b): wages, marginal products, and skill prices

Regard Q as a function of the aggregates $\mathcal{A} = \sum A_i$, $\mathcal{M} = \sum M_i$, $\mathcal{B} = \sum A_i M_i$, with marginal products (“skill prices”)

$$p_A = \frac{\partial Q}{\partial \mathcal{A}} = \frac{\alpha Q}{\mathcal{A}}, \quad p_M = \frac{\partial Q}{\partial \mathcal{M}} = \frac{\beta Q}{\mathcal{M}}, \quad p_B = \frac{\partial Q}{\partial \mathcal{B}} = \frac{(1 - \alpha - \beta)Q}{\mathcal{B}}.$$

A competitive firm pays each worker their marginal contribution. Adding one worker of type (a, m) raises $\mathcal{A}$ by a , $\mathcal{M}$ by m , and $\mathcal{B}$ by am , so

$$w(a, m) = \frac{\partial Q}{\partial H \text{ or } N} = p_A a + p_M m + p_B (a m). \quad (10)$$

Concretely $w_H = 2p_A + p_M + 2p_B$ and $w_N = p_A + 3p_M + 3p_B$.

Case $\alpha + \beta = 1$ ($p_B = 0$). The interaction term disappears and

$$w(a, m) = p_A a + p_M m$$

is *linear* in the worker's skills with prices p_A, p_M common to all workers. **Skill prices are well defined:** a worker's wage is the sum of the quantities of each skill she carries valued at economy-wide prices, and a hedonic regression of wages on (A_i, M_i) recovers them.

Case $\alpha + \beta < 1$ ($p_B > 0$). Now

$$w(a, m) = p_A a + p_M m + p_B (a m),$$

which is *non-separable*: the marginal return to ability is $\partial w / \partial a = p_A + p_B m$, i.e. it depends on how much motivation the *same* worker also supplies. **A single price of “a unit of ability” does not exist:** it is complementary with the worker's other skills.

Usefulness of skill prices. The notion is useful precisely when the aggregate skill quantities are sufficient statistics for output ($\alpha + \beta = 1$, no within-worker complementarity): then wages are linear in characteristics, hedonic coefficients are interpretable, and statements like “the price of cognitive skill rose” (the SBTC narrative) are meaningful. Once skills are bundled and complementary within individuals ($\alpha + \beta < 1$, the $\sum A_i M_i$ term), the implicit price of a characteristic is worker-specific, no common skill price exists, and a linear hedonic regression misidentifies it. This is Rosen's (1983) and Heckman-Scheinkman's (1987) warning about *bundling*: aggregation into “skill prices” is legitimate only under strong separability of the technology.

Question 3: On-the-Job Search

Time is discrete (months), agents are infinitely lived and risk neutral, and discount at $\frac{1}{1+r}$. Flow utility is $w - c$ when employed at w and b when searching; offers arrive at rate λ in *both* states, jobs are destroyed at rate δ , and offers are drawn from F (density f). Wages and benefits are paid at the end of the month, hence discounted.

Part (a): reading equation (1)

$$U(w) = \underbrace{\frac{w-c}{1+r}}_{(i)} + \underbrace{\frac{1}{1+r} \lambda (1 - F(w))}_{(ii)} \left[\delta V + (1 - \delta) \int_w^\infty U(\omega) \frac{f(\omega)}{1 - F(\omega)} d\omega \right] + \underbrace{\frac{1}{1+r} (1 - \lambda (1 - F(w)))}_{(iii)} [\delta V + (1 - \delta) U(w)]$$

- (i) End-of-month flow payoff of the current job, $w - c$, discounted one period.
- (ii) With probability λ an offer arrives, and it is worth moving for only if it beats the current wage, i.e. with probability $1 - F(w)$; the offer ω is then a draw from F *truncated* to $\omega > w$ (density $f(\omega)/(1 - F(w))$). But the new job must survive into next period: with probability δ it is destroyed and the worker falls into unemployment V , with probability $1 - \delta$ she enjoys $U(\omega)$.

- (iii) With the complementary probability she has no acceptable offer (no offer, or an offer below w) and stays put; entering next period her *current* job is destroyed with probability δ ($\rightarrow V$) or survives ($\rightarrow U(w)$).

Collecting terms (the δV pieces have total weight one) gives the compact form

$$U(w) = \frac{w - c}{1 + r} + \frac{\delta V}{1 + r} + \frac{1 - \delta}{1 + r} \left[\lambda \int_w^\infty U(\omega) f(\omega) d\omega + (1 - \lambda(1 - F(w))) U(w) \right]. \quad (11)$$

Part (b): the value of unemployment

An unemployed worker earns b (end of month, discounted), and with probability λ draws an offer that she accepts iff $\omega \geq w^*$. An accepted job again faces destruction δ next period; otherwise she remains unemployed with continuation V :

$$V = \frac{b}{1 + r} + \frac{1}{1 + r} \left[\lambda \int_{w^*}^\infty (\delta V + (1 - \delta) U(\omega)) f(\omega) d\omega + (1 - \lambda(1 - F(w^*))) V \right]. \quad (12)$$

Part (c): the reservation wage is $w^* = b + c$

The reservation wage solves $U(w^*) = V$. Evaluate (11) at w^* , set $U(w^*) = V$, and write $q \equiv \lambda(1 - F(w^*))$ and $J \equiv \lambda \int_{w^*}^\infty U(\omega) f(\omega) d\omega$:

$$(1 + r)V = (w^* - c) + \delta V + (1 - \delta)[J + (1 - q)V].$$

Do the same to (12):

$$(1 + r)V = b + \delta qV + (1 - \delta)J + (1 - q)V.$$

Subtract. The integral J cancels, and the two continuation blocks are *identical*: $\delta V + (1 - \delta)(1 - q)V = V(1 - q + \delta q)$ on one side and $\delta qV + (1 - q)V = V(1 - q + \delta q)$ on the other. What remains is

$$w^* - c = b \implies \boxed{w^* = b + c.}$$

Intuition and contrast with the no-OJS model. Search is equally productive while employed and while unemployed (λ is common to both states), and the destruction risk δ is the same. Hence the *option value of future search* and the job-loss risk are identical at $U(w^*)$ and at V , and they cancel: the worker accepts as soon as the flow payoff of working, $w - c$, covers the flow payoff of unemployment, b . In the basic McCall model *without* on-the-job search, accepting a job forecloses further search (employment is absorbing), so the reservation wage embeds a strictly positive option value: w^* exceeds b (+disutility) by the expected gain from sampling better offers, $w^* = b - \nu + \frac{1}{r} E[w - w^* | w \geq w^*](1 - F(w^*))$, and the worker is “picky.” With OJS that motive disappears because she keeps the search option after accepting, collapsing the reservation wage to the static flow comparison $w^* = b + c$.

Part (d): duration, exit wage, and UI policy

Each month unemployed the worker exits with probability $q = \lambda(1 - F(w^*))$, so unemployment duration is geometric:

$$E[\text{duration}] = \frac{1}{\lambda(1 - F(b + c))}, \quad E[\text{exit wage}] = E[w | \omega \geq w^*] = \frac{\int_{w^*}^\infty \omega f(\omega) d\omega}{1 - F(w^*)}. \quad (13)$$

Since $w^* = b + c$ is increasing in b , a more generous benefit raises both:

$$\frac{\partial E[\text{duration}]}{\partial b} = \frac{f(w^*)}{\lambda(1 - F(w^*))^2} > 0, \quad \frac{\partial E[\text{exit wage}]}{\partial b} = \frac{f(w^*)}{1 - F(w^*)} (E[\omega | \omega \geq w^*] - w^*) > 0.$$

Higher b makes workers choosier, lengthening joblessness but improving the accepted wage. At the calibration of part (g) ($w^* = 800$): $1 - F(w^*) = 0.841$, the monthly exit hazard is 0.168, expected duration is 5.94 months, and the expected re-employment wage is 1,314.

Policy. This is precisely the tension stressed in class around extending UI (Nekoei-Weber 2017): longer potential benefits raise reservation wages, increasing both unemployment duration (a moral-hazard / fiscal cost) *and* re-employment wages (a match-quality benefit). The model rationalizes the empirical finding that UI extensions can *raise* accepted wages through selectivity, so the welfare verdict hinges on trading off longer joblessness against better matches rather than on duration alone.

Part (e): $U(\bar{w})$ and $U(w^*)$ on support $[0, \bar{w}]$

At the top wage no better offer can arrive ($1 - F(\bar{w}) = 0$), so the search terms in (11) vanish:

$$U(\bar{w}) = \frac{\bar{w} - c}{1 + r} + \frac{\delta V}{1 + r} + \frac{1 - \delta}{1 + r} U(\bar{w}) \implies \boxed{U(\bar{w}) = \frac{\bar{w} - c + \delta V}{r + \delta}}.$$

Intuitively, the worker at $\bar{w}$ has no upgrade option, so she simply holds a job that yields flow $\bar{w} - c$ and dies at rate δ : the flow is capitalized at the *effective* discount rate $r + \delta$, plus the value δV of the job-loss contingency.

Using $V = U(w^*)$ in the contraction (14) below at $w = w^*$ gives

$$\boxed{U(w^*) = V = \frac{b}{r} + \frac{\lambda(1 - \delta)}{r} \int_{w^*}^{\bar{w}} (U(\omega) - V) f(\omega) d\omega}.$$

The unemployed worker's value is the perpetuity value of the benefit, b/r , plus the *option value of search*: the expected capital gain from climbing to better jobs, $U(\omega) - V$, weighted by the arrival rate and survival probability. (Numerically $V = 49,292$, well above $b/r = 40,000$.)

Part (f): the slope $U'(w)$ and the picture

Differentiate (14). Writing $I(w) = \int_w^{\bar{w}} (U(\omega) - U(w)) f(\omega) d\omega$ one has $I'(w) = -U'(w)(1 - F(w))$, so

$$U'(w) = \frac{1}{r + \delta} - \frac{\lambda(1 - \delta)}{r + \delta} U'(w)(1 - F(w)) \implies \boxed{U'(w) = \frac{1}{r + \delta + \lambda(1 - \delta)(1 - F(w))} > 0}.$$

Because $1 - F(w)$ falls as w rises, the denominator falls and U' rises: U is **increasing and convex**. At the top, $U'(\bar{w}) = \frac{1}{r + \delta}$ (the numerical derivative gives $3.7037 = 1/0.27$, matching exactly). The slope is flattest at the bottom, $U'(0) = 1/[r + \delta + \lambda(1 - \delta)]$, because there the option to upgrade is most valuable, so a marginal wage increase is worth comparatively little.

Figure 1 plots the solution: $U(w)$ is increasing and convex, while $V = U(w^*)$ is a horizontal line crossing U at $w^* = b + c = 800$. For $w < w^*$, $U(w) < V$ (the worker rejects); for $w > w^*$, $U(w) > V$ (she accepts).

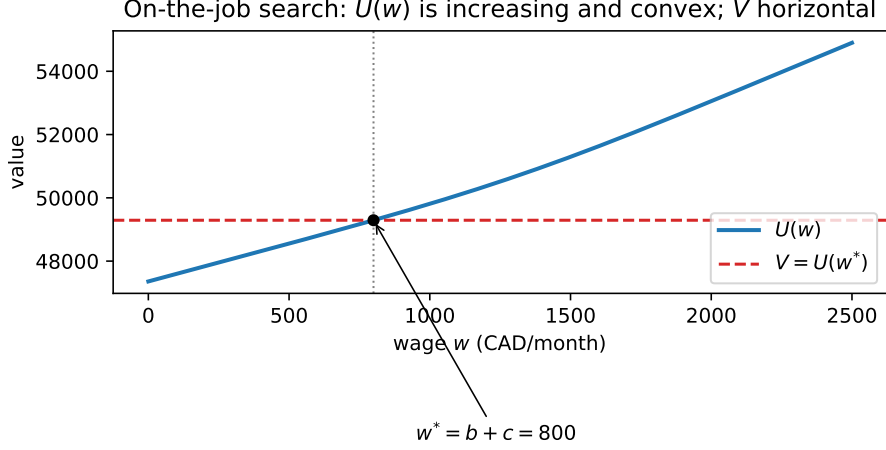


Figure 1: Numerical solution of the on-the-job-search model ($b = 800$, $c = 0$, $\lambda = 0.2$, $r = 0.02$, $\delta = 0.25$, $F = N(1200, 400)$, $\bar{w} = 2500$). $U(w)$ increasing and convex; $V = U(w^*)$ horizontal; they cross at $w^* = b + c = 800$.

Part (g): from (1) to the contraction, and the numerical solution

Substitute $\int_w^{\bar{w}} U(\omega) f d\omega = \int_w^{\bar{w}} (U(\omega) - U(w)) f d\omega + U(w)(1 - F(w))$ into (11):

$$U(w) = \frac{w - c}{1 + r} + \frac{\delta V}{1 + r} + \frac{1 - \delta}{1 + r} U(w) + \frac{\lambda(1 - \delta)}{1 + r} \int_w^{\bar{w}} (U(\omega) - U(w)) f(\omega) d\omega.$$

Move $\frac{1 - \delta}{1 + r} U(w)$ to the left (the coefficient becomes $\frac{r + \delta}{1 + r}$), multiply through by $\frac{1 + r}{r + \delta}$, and use $V = U(b + c)$:

$$\boxed{U(w) = \frac{w - c}{r + \delta} + \frac{\delta U(b + c)}{r + \delta} + \frac{\lambda(1 - \delta)}{r + \delta} \int_w^{\bar{w}} (U(\omega) - U(w)) f(\omega) d\omega} \quad (14)$$

which is the required functional equation $U = T\{U\}$.

Existence/uniqueness. The algebraically equivalent Bellman map (11) places total weight $\frac{\delta}{1 + r} + \frac{1 - \delta}{1 + r} [\lambda(1 - F) + (1 - \lambda(1 - F))] = \frac{1}{1 + r} < 1$ on the unknown U , so T is a contraction of modulus $\frac{1}{1 + r}$ and has a unique fixed point; successive approximation converges.

Algorithm. Discretize $[0, 2500]$ on a grid of step 1 (so $w^* = 800$ is a node), evaluate f from $N(1200, 400)$, and iterate $U^{n+1} = T\{U^n\}$ on (14) from $U^1(w) = (w - c)/(r + \delta)$ until $\|U^{n+1} - U^n\|_\infty < 10^{-9}$, reading $V = U(b + c)$ each pass. The code is in `ojs_solve.py`. It converges in 376 iterations and yields

$$V = U(800) = 49,291.6, \quad U(0) = 47,357.3, \quad U(2500) = 54,899.7.$$

Three internal checks confirm the solution: (i) the closed form $U(\bar{w}) = (\bar{w} - c + \delta V)/(r + \delta)$ reproduces $U(2500)$ to the penny; (ii) iterating the original Bellman form (11) returns the same V to within 4×10^{-8} ; (iii) the numerical $U'(\bar{w}) = 3.7037$ equals $1/(r + \delta)$.

Question 4: Monopsony and Wage Posting

A monopsonist faces firm-level supply $L(w)$, linear technology, and output price P , with profit $\pi(w) = PL(w) - wL(w) = (P - w)L(w)$ (so the marginal revenue product of labor is the constant P).

Part (a): first-order condition

$$\pi'(w) = -L(w) + (P - w)L'(w) = 0 \implies \boxed{P - w = \frac{L(w)}{L'(w)}} \iff \frac{P - w}{w} = \frac{1}{\varepsilon_L}, \quad w = \frac{\varepsilon_L}{1 + \varepsilon_L} P,$$

where $\varepsilon_L = \frac{L'(w)w}{L(w)}$ is the firm's labor-supply elasticity. The wage is marked down below the marginal product P by the inverse elasticity, the monopsony wedge.

Part (b): a supply curve that makes profit wage-insensitive

Require $\pi(w) = (P - w)L(w) = K$ constant. Then

$$\boxed{L(w) = \frac{K}{P - w}, \quad w < P, \quad K > 0.} \quad (15)$$

Indeed $\pi(w) = (P - w)\frac{K}{P - w} = K$ for every $w < P$, and one checks $(P - w)L'(w) = (P - w)\frac{K}{(P - w)^2} = \frac{K}{P - w} = L(w)$, so the first-order condition holds at *every* wage: the firm is indifferent across the whole range, the static analogue of Burdett-Mortensen wage dispersion.

Part (c): the implied elasticity

$$\varepsilon_L(w) = \frac{L'(w)w}{L(w)} = \frac{w/(P - w)^2}{1/(P - w)} = \boxed{\frac{w}{P - w}},$$

which is *increasing* in w : $\varepsilon_L \rightarrow 0$ as $w \rightarrow 0$ and $\varepsilon_L \rightarrow \infty$ as $w \rightarrow P$. Higher-wage firms face more elastic supply (less monopsony power), which is directionally plausible. But the specific form is knife-edge: it requires the elasticity to explode exactly as $w \rightarrow P$, whereas the empirical literature (e.g. Dube et al. discussed in class) finds finite, roughly *constant* firm-level elasticities of order a few. A constant (isoelastic) supply curve, the usual empirical benchmark, would *not* deliver wage-insensitive profits. So static wage posting reproduces Burdett-Mortensen indifference only under a non-robust supply schedule; the genuine BM result rests on dynamic equilibrium forces, not a single firm's static supply curve.

Part (d): inverse supply, MFC, demand, and multiplicity

Inverting (b), $P - w = K/L$, so

$$w(L) = P - \frac{K}{L} \quad (\text{increasing, concave, } \rightarrow P \text{ as } L \rightarrow \infty),$$

$$\text{MFC} = \frac{d}{dL} [w(L)L] = \frac{d}{dL} [PL - K] = P.$$

The marginal factor cost is the *constant* P , and with linear technology the inverse labor demand (MRPL) is also the constant P . Hence **MFC and labor demand coincide** (both horizontal at P), while the inverse supply curve rises toward P from below (Figure 2). Because the firm's marginal benefit of hiring (P) equals its marginal cost (P) at *every* employment level, the optimality condition $MFC = MRPL$ holds everywhere: every point on the supply curve is an equilibrium, each delivering the same profit K . The coincidence of the MFC and demand schedules, rather than a unique crossing, is what generates the continuum of equilibria.

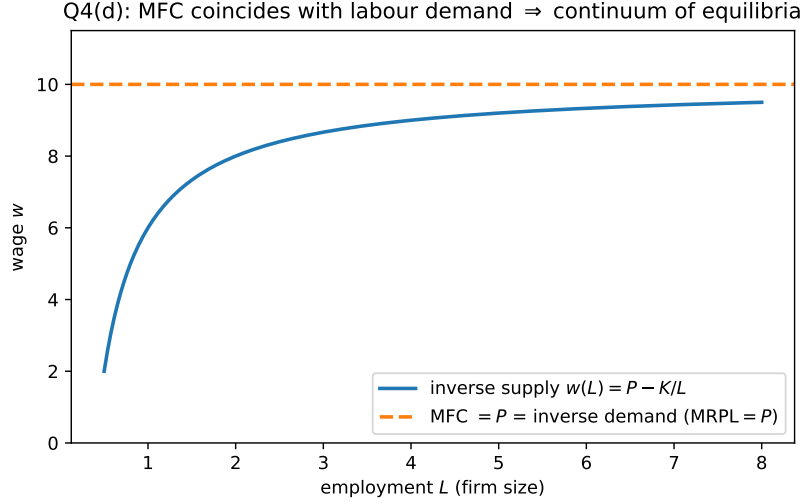


Figure 2: Inverse supply $w(L) = P - K/L$ rises toward P ; $MFC = P$ coincides with inverse labor demand $MRPL = P$. Their overlap (not a unique intersection) generates a continuum of equilibria.

Part (e): a minimum wage at $P/2$

Start at $w^* < P/2$ with employment $L^* = K/(P - w^*)$. Imposing $w_{\min} = P/2$ (still below P , so profit remains K and the firm complies at exactly $P/2$) moves the firm *up* its supply curve to

$$L' = \frac{K}{P - P/2} = \frac{2K}{P} > \frac{K}{P - w^*} = L^* \quad (\text{since } w^* < P/2 \Rightarrow P - w^* > P/2).$$

Employment *rises*. This is the classic monopsony result: a minimum wage set between the posted wage and the marginal product raises wages *and* employment by moving the firm along an upward-sloping supply curve toward the competitive point $w = P$ (Figure 3; at $P = 10$, $K = 4$, employment climbs from 0.571 to 0.800).

Question 5: Firm Pay, Amenities, and Compensating Differentials

Worker utility at firm j is $V_j = \log \psi_j + \phi \log a_j$ ($\phi > 0$); profit is $\pi_j = p_j - c_j a_j - \psi_j$; and the firm must deliver at least $\bar{V}_j$. Since pay and amenities are costly, the constraint binds, $\log \psi_j + \phi \log a_j = \bar{V}_j$.



Figure 3: A minimum wage at $P/2$ moves the firm up the supply curve from (L^*, w^*) to $(L', P/2)$: employment increases.

Part (a): optimal pay and amenities

The firm minimizes the cost $\psi_j + c_j a_j$ of delivering $\bar{V}_j$. Tangency of the iso-utility curve to the iso-cost line gives $\frac{1/\psi_j}{\phi/a_j} = \frac{1}{c_j}$, i.e. $c_j a_j = \phi \psi_j$. Combining with the binding constraint,

$$\log \psi_j = \frac{1}{1+\phi} \bar{V}_j + \frac{\phi}{1+\phi} \log c_j - \frac{\phi}{1+\phi} \log \phi \quad (16)$$

$$\log a_j = \frac{1}{1+\phi} \bar{V}_j - \frac{1}{1+\phi} \log c_j + \frac{1}{1+\phi} \log \phi \quad (17)$$

(One verifies $\log \psi_j + \phi \log a_j = \bar{V}_j$.) Note that productivity p_j does not enter: it is additively separable in profit and so does not affect the pay/amenity mix chosen to hit $\bar{V}_j$.

Part (b): common utility $\bar{V}_j = \bar{V}$

With $\bar{V}_j$ fixed, only c_j varies: $\log \psi_j = \text{const} + \frac{\phi}{1+\phi} \log c_j$ and $\log a_j = \text{const} - \frac{1}{1+\phi} \log c_j$. They are deterministic, oppositely-signed functions of the single driver $\log c_j$, so

$$\text{corr}(\log \psi_j, \log a_j) = -1.$$

Intuition. Holding utility fixed, firms differ only in how cheaply they can supply amenities. High- c_j firms deliver utility through pay (high ψ , low a); low- c_j firms through amenities. This is the pure *compensating-differentials* trade-off: along an iso-utility locus pay and amenities substitute one-for-one, producing a perfect negative correlation.

Part (c): common amenity cost $c_j = \bar{c}$

Now only $\bar{V}_j$ varies: $\log \psi_j = \frac{1}{1+\phi} \bar{V}_j + \text{const}$ and $\log a_j = \frac{1}{1+\phi} \bar{V}_j + \text{const}$, both increasing in $\bar{V}_j$, so

$$\boxed{\text{corr}(\log \psi_j, \log a_j) = +1.}$$

Intuition. When the cost of amenities is common but firms differ in the utility they deliver, high- $\bar{V}$ firms are simply “better”: they pay more *and* offer better amenities. This is *vertical differentiation*, good firms versus bad firms, and produces a positive pay-amenity correlation, the opposite of compensating differentials.

Part (d): the general covariance

Writing $A = \frac{1}{1+\phi}$, $\log \psi_j = A\bar{V}_j + \phi A \log c_j + \text{const}$ and $\log a_j = A\bar{V}_j - A \log c_j + \text{const}$, so

$$\boxed{\text{cov}(\log \psi_j, \log a_j) = \frac{1}{(1+\phi)^2} \left[\text{var}(\bar{V}_j) + (\phi - 1) \text{cov}(\bar{V}_j, \log c_j) - \phi \text{var}(\log c_j) \right].} \quad (18)$$

This nests (b) ($\text{var}(\bar{V}) = 0 \Rightarrow \text{cov} = -\phi A^2 \text{var}(\log c) < 0$) and (c) ($\text{var}(\log c) = 0 \Rightarrow \text{cov} = A^2 \text{var}(\bar{V}) > 0$). **It can be positive:** whenever firm heterogeneity is dominated by differences in deliverable utility $\bar{V}_j$ (vertical differentiation) rather than in amenity costs c_j . The sign is a horse race between the Rosen force (variation in c_j , pushing toward negative correlation) and the firm-quality force (variation in $\bar{V}_j$, pushing positive). A positive measured covariance, namely the empirical pattern that better-paying firms also have better amenities (the “no compensating differentials” finding shown in class), therefore does *not* reject compensating differentials; it merely says that firm-quality dispersion outweighs amenity-cost dispersion.

Part (e): regressing $\log \psi_j$ on $\bar{V}_j$

$$\text{slope} = \frac{\text{cov}(\log \psi_j, \bar{V}_j)}{\text{var}(\bar{V}_j)} = \frac{1}{1+\phi} \left[1 + \phi \frac{\text{cov}(\bar{V}_j, \log c_j)}{\text{var}(\bar{V}_j)} \right].$$

It identifies the structural parameter $\frac{1}{1+\phi}$ (hence ϕ) *only* under the orthogonality assumption $\text{cov}(\bar{V}_j, \log c_j) = 0$: firms that deliver more utility are not systematically cheaper/dearer at providing amenities. Otherwise the omitted $\log c_j$ contaminates the slope (omitted-variable bias) and it identifies no single parameter.

Part (f): the residual variance and “Rosen motives”

Let $\hat{e}_j$ be the residual from the regression in (e). Then

$$\text{var}(\hat{e}_j) = \text{var}(\log \psi_j) - \text{cov}(\log \psi_j, \bar{V}_j)^2 / \text{var}(\bar{V}_j),$$

which after substitution collapses to

$$\boxed{\text{var}(\hat{e}_j) = \frac{\phi^2}{(1+\phi)^2} \left[\text{var}(\log c_j) - \frac{\text{cov}(\bar{V}_j, \log c_j)^2}{\text{var}(\bar{V}_j)} \right] = \frac{\phi^2}{(1+\phi)^2} \text{var}(\log c_j | \bar{V}_j).} \quad (19)$$

Interpretation. The residual is the part of firm pay *not* explained by the firm’s overall utility provision $\bar{V}_j$: it is the pay variation driven by differences in the cost of amenities c_j (orthogonal to $\bar{V}_j$). Firms with high amenity costs pay more to compensate, for reasons unrelated to how “good” the firm is. This is exactly the compensating-differential, or **Rosen-motive**, component of firm pay.

Can one identify the Rosen-motive share? The share of firm-pay variance due to Rosen motives is $\text{var}(\hat{e}_j)/\text{var}(\log \psi_j)$. Computing it requires (i) data on $\bar{V}_j$ (unobserved, but exactly what Sorkin (2018) recovers from job-to-job flows via his PageRank/revealed-preference procedure) and (ii) the orthogonality assumption $\text{cov}(\bar{V}_j, \log c_j) = 0$, under which firm-pay variance splits cleanly into a vertical-quality piece (explained by $\bar{V}_j$) and a compensating-differential piece (the residual). This is the variance decomposition behind Sorkin’s conclusion that roughly half of firm-pay variation reflects compensating differentials. If instead $\text{cov}(\bar{V}_j, \log c_j) \neq 0$, the residual captures only the component of amenity-cost variation orthogonal to firm utility, so $\text{var}(\hat{e}_j)$ is a *lower bound* on the true Rosen-motive contribution.